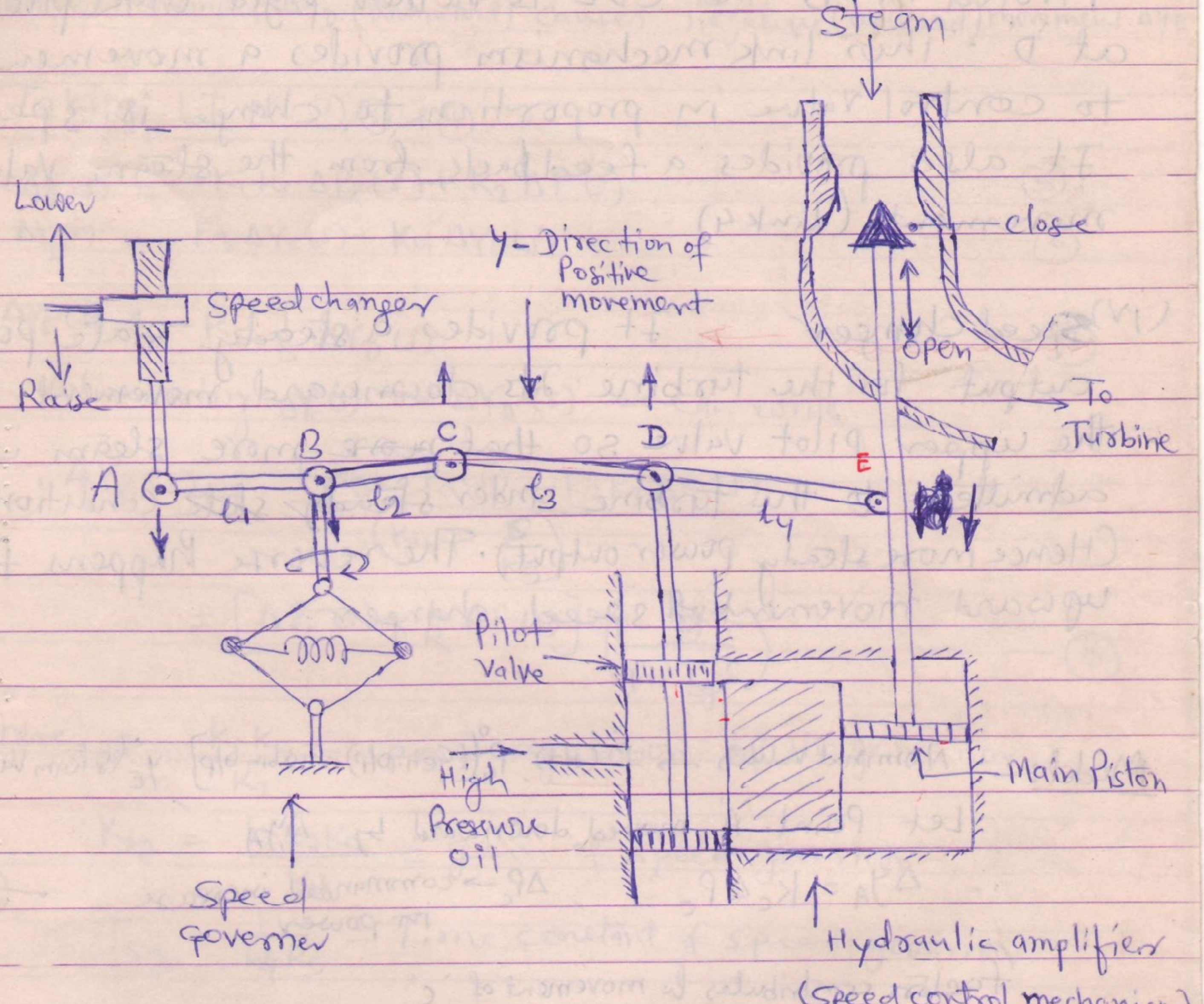
Assigment-3

ANS_Q1 Turbine speed Governing system :—



(1) Fly ball speed Governer :— It senses the change in speed (frequency). As the speed increases the fly balls move outwards and point B on linkage mechanism moves downwards. The reverse happens when the speed decreases.

(2) Hydraulic amplifier → It comprises a pilot valve and main piston arrangement. Low power level pilot valve movement is converted into high power level piston valve movement.

against hight pressure steam.

(iii) Linkage mechanism :- ABC is a rigid link pivoted at B and CDE is another rigid link pivoted at 'D'. This link mechanism provides a movement to control valve in proportion to change in speed. It also provides a feed back from the steam value movement (link 4)

(iv) Speed changer → It provides a steady state power output for the turbine. Its downward movement opens the upper pilot valve so that ~~more~~ more steam is admitted to the turbine under steady ~~state~~ conditions (Hence more steady power output). The reverse happens for upward movement of speed changer.

Model:- Nominal Values → f° (Hz), P_G° [(Gen. o/p) = Turbin o/p] Y_E° (Steam Valve setting)

Let Point 'A' moved downward by Δy_A

$\therefore \quad \Delta y_A = K_c \Delta P_c$, $\Delta P_c \rightarrow$ commanded increase in power ——(1)

factors contributes to movement of 'C'

(i) Δy_A contributes $-\left(\frac{l_2}{l_1}\right)\Delta y_A$ or $-K_1 \Delta y_A$ (i.e. upward or $-K_1 K_c \Delta P_c$

(ii) Increase in freq. Δf causes the flyballs to move outwards so that B moves downwards by proportional amount $K_2' \Delta f$. The consequent movement of 'C' with 'A' remaining fixed at Δy_A is $+\left(\frac{l_1+l_2}{l_1}\right) K_2' \Delta f$

$= +K_2 \Delta f$ (i.e. downward)

Net movement of 'C'

$\Delta y_c = -K_1 K_c \Delta P_C + K_2 \Delta f$ ——(2)

movement of 'D'

$\Delta y_D = \left(\frac{l_4}{l_3+l_4}\right)\Delta y_c + \left(\frac{l_3}{l_3+l_4}\right)\Delta y_E = K_3 \Delta y_c + K_4 \Delta y_E$ ——(3)